

Ordnance is disposed of by focusing high-power laser beam on the ordnance casing, thereby heating it until temperature of the backplane of the casing exceeds ignition temperature of the explosive filler. The explosive filler ignites and begins to burn (Fig. 12), and the process is independent of type of fusing used by target explosive. This leads to a low-level detonation or deflagration rather than full-power detonation.

Advantages of the use of laser energy for ordnance disposal include large magazine, high precision, controllable effects with reduced collateral damage and assured and fast disposal from safe stand-off ranges.

Application scenarios

The laser ordnance neutralisation system is being considered for a range of application scenarios. One possible application relevant to homeland security is the disposal of IEDs, also called roadside bombs. This system offers a quick, safe and reliable method of neutralising these roadside bombs from a safe stand-off distance in excess of 150 metres.

Another possible application is in the disposal of friendly explosives that have outlived their shelf lives. Since full-scale conventional wars are not fought every now and then, there is a large inventory of ammunition stored in ammunition depots having exceeded their shelf lives and waiting to be neutralised.

Neutralisation of such large quantities of ammunition is not only a cumbersome exercise, it poses a safety hazard, too. Laser-driven disposal of UXOs is not only safe, it also allows rapid neutralisation of large quantities of explosive materials.

Representative laser ordnance neutralisation systems

There can be various possible variants of the ordnance neutralisation



Fig. 12: Explosive deflagration



Fig. 13: ZEUS-HLONS laser ordnance neutralisation system



Fig. 14: Laser Avenger system

tion system. One configuration is the standalone high-power laser system mounted on a vehicular platform. ZEUS-HLONS and LASER AVENGER are the examples of this category.

In another possible variant, the system may be mounted on a remotely-controlled vehicle with a robotic arm. While the laser system along with its power and thermal management system could be mounted on the platform, the laser delivery system could be mounted on the robotic arm.

In addition to these configurations, high-power laser system may be integrated with other means such as ground penetration radar for the detection of buried

explosives. The integrated system allows detecting and neutralising both surface-laid as well as buried explosives.

In some cases, the laser ordnance disposal system is mounted on a remotely-controlled weapon station to get both defensive as well as offensive capability. Rafael's high-energy laser weapon system called THOR, developed in Israel to defeat IEDs, roadside bombs, UXOs and other potentially hazardous explosives, is an example of this type of system.

ZEUS-HLONS (HUMMWV laser ordnance neutralisation system) of the USA (Fig. 13) is a vehicle-mounted laser ordnance neutralisation system. The concept of neutralisation of live ordnances using laser energy was first demonstrated in the field in 1994 with the development and field testing of Mobile Ordnance Disrupter System (MODS) that employed a 1.1kW-arc-lamp driven solid-state laser mounted on an M113 A2 armoured personnel carrier.

ZEUS system initially employed a 500-watt laser. Latest version of ZEUS system is called ZEUS-II. It consists of a multi-kilowatt-class laser, beam director, operator's station and all support sub-systems on a single vehicle. The system can be integrated with M1114 HMMWV, M-ATV or other MRAP vehicles.

ZEUS has been extensively field evaluated against more than 40 different types of ordnance, which include landmines, improved conventional munitions, mortar rounds, rifle grenades, rockets and artillery projectiles ranging from small, plastic landmines to large, thick-walled 155mm projectiles and 227kg general-purpose bombs. The system has demonstrated its effectiveness in a variety of missions including counter-IED operations, clearing land mines, clearing UXOs from battlefields or, during peace-keeping missions, clearing active